



New Wheels Project

Data Querying and Analytics with SQL

Solution Document

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Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

shipper_id: Unique ID of the Shipper
shipper_name: Name of the Shipper
shipper_contact_details: Contact detail of the Shipper
product_id: Unique ID of the Product
vehicle_maker: Vehicle Manufacturing company name
vehicle_model: Vehicle model name
vehicle_color: Color of the Vehicle
vehicle_model_year: Year of Manufacturing
vehicle_price: Price of the Vehicle
quantity: Ordered Quantity
customer_id: Unique ID of the customer
customer_name: Name of the customer
gender: Gender of the customer
job_title: Job Title of the customer
phone_number: Contact detail of the customer
email_address: Email address of the customer
city: Residing city of the customer
country: Residing country of the customer
state: Residing state of the customer
customer_address: Address of the customer
order_date: Date on which customer ordered the vehicle
order_id: Unique ID of the order
ship_date: Shipment Date
ship_mode: Shipping Mode/Class
shipping: Shipping Ways
postal_code: Postal Code of the customer
discount: Discount given to the customer for the particular order by credit card in percentage
credit_card_type: Credit Card Type
credit_card_number: Credit card number
customer_feedback: Feedback of the customer
quarter_number : Quarter Number

Entity-Relationship Diagram

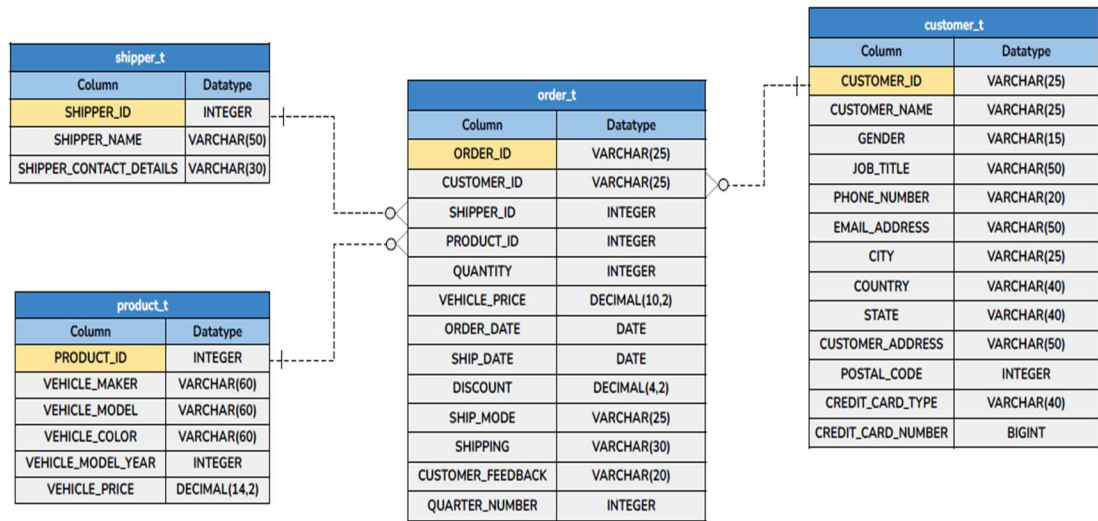


Figure 1 ER diagram

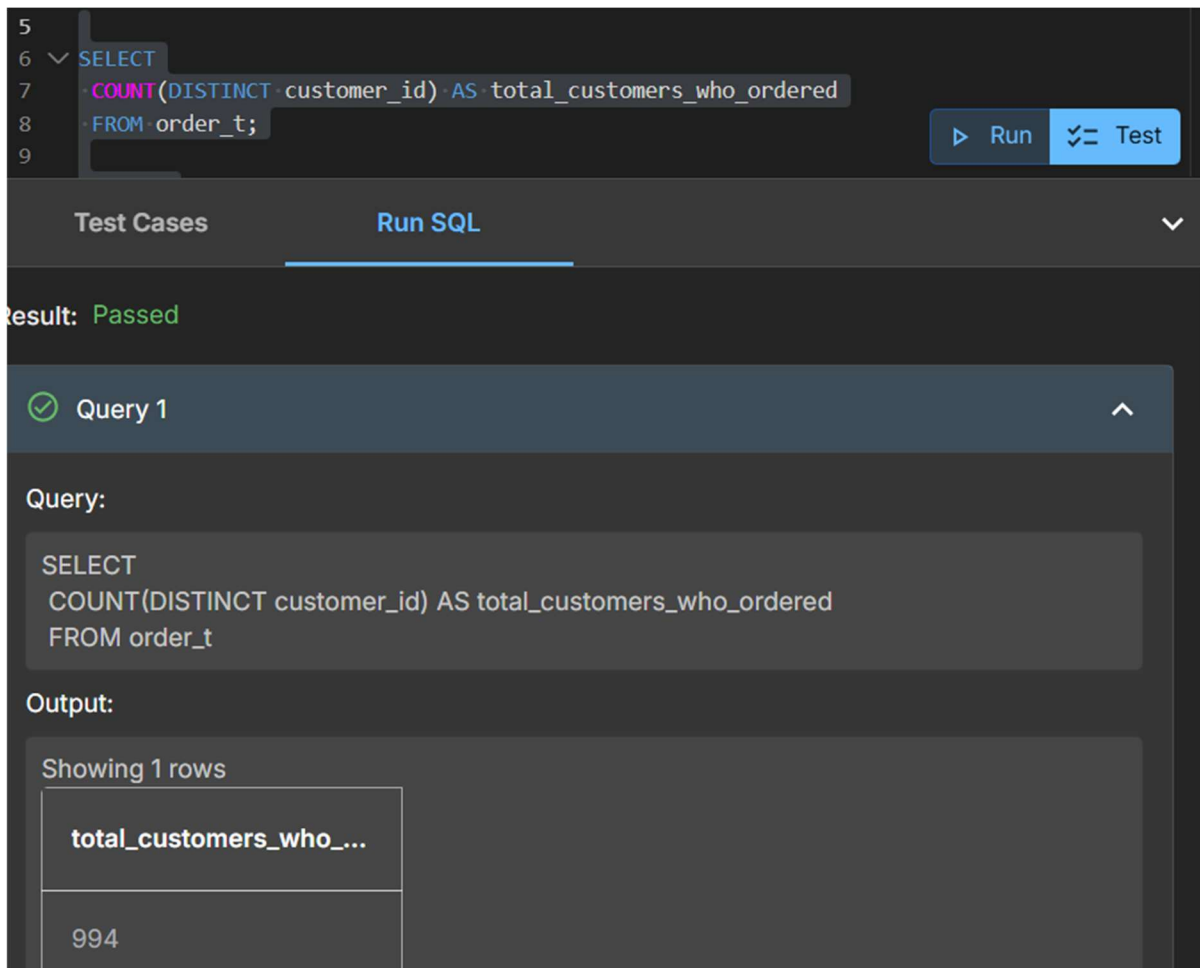
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
COUNT(DISTINCT customer_id) AS total_customers_who_ordered
FROM order_t;
```

```
SELECT
c.state,
COUNT(DISTINCT c.customer_id) AS number_of_customers
FROM customer_t c
JOIN order_t o
ON c.customer_id = o.customer_id
GROUP BY c.state;
```

Output:



5
6 SELECT
7 COUNT(DISTINCT customer_id) AS total_customers_who_ordered
8 FROM order_t;
9

Run Test

Test Cases Run SQL

Result: Passed

Query 1

Query:

```
SELECT
COUNT(DISTINCT customer_id) AS total_customers_who_ordered
FROM order_t
```

Output:

Showing 1 rows

total_customers_who_...
994

Figure 2 Question1 Output Screenshots

```
10 SELECT
11     c.state,
12     COUNT(DISTINCT c.customer_id) AS number_of_customers
13 FROM customer_t c
14 JOIN order_t o
15     ON c.customer_id = o.customer_id
16 GROUP BY c.state;
```

▶ Run ⚙ Test

Test Cases Run SQL

Result: **Passed**

✓ Query 1

✓ Query 2

Query:

```
SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS number_of_customers
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.state
```

Output:

Test Cases	Run SQL
Output:	
Showing first 10 rows out of 49 rows	
state	number_of_customers
Alabama	29
Alaska	10
Arizona	26
Arkansas	6
California	97
Colorado	33
Connecticut	22

Table 1 Question 1 Output

state	number_of_customers
Alabama	29
Alaska	10
Arizona	26
Arkansas	6
California	97
Colorado	33
Connecticut	22
Delaware	6
District of Columbia	35
Florida	86
Georgia	18
Hawaii	6
Idaho	7
Illinois	25
Indiana	21
Iowa	11
Kansas	13
Kentucky	8
Louisiana	20

Maine	1
Maryland	14
Massachusetts	14
Michigan	17
Minnesota	17
Mississippi	2
Missouri	23
Montana	3
Nebraska	7
Nevada	17
New Hampshire	3
New Jersey	9
New Mexico	5
New York	69
North Carolina	20
North Dakota	2
Ohio	33
Oklahoma	16
Oregon	7
Pennsylvania	25
South Carolina	9
Tennessee	23
Texas	97
Utah	10
Vermont	1
Virginia	24
Washington	28
West Virginia	10
Wisconsin	8
Wyoming	1

Observations and Insights:

- Total Customer Base: There are 994 unique customers who placed orders.
- Average Presence: On average, there are approximately 20 customers per state.
- **California** and **Texas** are the clear leaders, each with **97 customers**, followed closely by **Florida (86)** and **New York (69)**. These four states combined represent a large portion of the total customer base, indicating they are likely primary hubs for your business operations.
- **Maine, Vermont, and Wyoming** each have only **1 customer**, while **Mississippi and North Dakota** have only **2**. These regions may represent opportunities for growth through targeted marketing campaigns or potential areas to investigate for logistical challenges
- Distribution of customers across state is as in below bar chart.



Figure 3 Qution1 Chart

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(o.customer_id) AS count_customers
FROM order_t o
JOIN product_t p
    USING(product_id)
GROUP BY p.vehicle_maker
ORDER BY count_customers DESC
LIMIT 5;
```

Output:

Test Cases

Run SQL

Result: Passed

✓ Query 1

✓ Query 2

✓ Query 3

Query:

```
SELECT
    p.vehicle_maker,
    COUNT(o.customer_id) AS count_customers
FROM order_t o
JOIN product_t p
    USING(product_id)
GROUP BY p.vehicle_maker
ORDER BY count_customers DESC
LIMIT 5
```

Output:

```

20 -- HINT: For each vehicle make what is the count of the customers.
21 SELECT
22     p.vehicle_maker,

```

Test Cases **Run SQL**

```

GROUP BY p.vehicle_maker
ORDER BY count_customers DESC
LIMIT 5

```

Output:

Showing 5 rows

vehicle_maker	count_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Figure 4 Question 2 Output Screenshot

Table 2 Question 2 Output

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

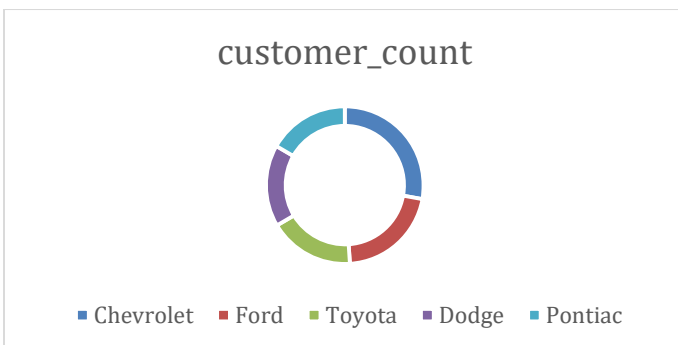


Figure 5 Question 2 Chart

Observations and Insights:

- **Chevrolet** and **Ford** are the clear market leaders, collectively accounting for nearly **49%** of the orders among these top 5 manufacturers. Chevrolet alone holds a significant lead with **83 customers**
- With Chevrolet and Ford representing nearly half of the top 5 volume, marketing campaigns or service packages tailored to these specific makes could offer high ROI (Return on Investment) due to the larger, concentrated user base.
- The high numbers for domestic brands (Chevrolet, Ford, Dodge) compared to international brands like Toyota may indicate a strong regional preference for domestic vehicle makers or better dealership/service network availability
- The consistency between the 3rd, 4th, and 5th positions suggests that any competitive advantage (e.g., better pricing, improved after-sales support, or loyalty incentives) could easily shift the rankings for the middle-tier manufacturers.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    state,
    vehicle_maker,
    order_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(o.order_id) AS order_count,
        RANK() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(o.order_id) DESC
        ) AS rnk
    FROM customer_t c
    JOIN order_t o ON c.customer_id = o.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY
        c.state,
        p.vehicle_maker
) AS ranked_data
WHERE rnk = 1;
```

Output:

```

33  -- Question 3:
34  -- Which is the most preferred vehicle maker in each state? [4 marks]
35  -- Hint: Use the window function RANK() to rank based on the count of customers for each
36  -- After ranking, take the vehicle maker whose rank is 1.
37
38  SELECT
39      state,
40      vehicle_maker,
41      order_count
42  FROM (
43      SELECT
44          c.state,
45          p.vehicle_maker,
46          COUNT(o.order_id) AS order_count,
47          RANK() OVER (
48              PARTITION BY c.state
49              ORDER BY COUNT(o.order_id) DESC

```

Run Test

Test Cases Run SQL

Result: Passed

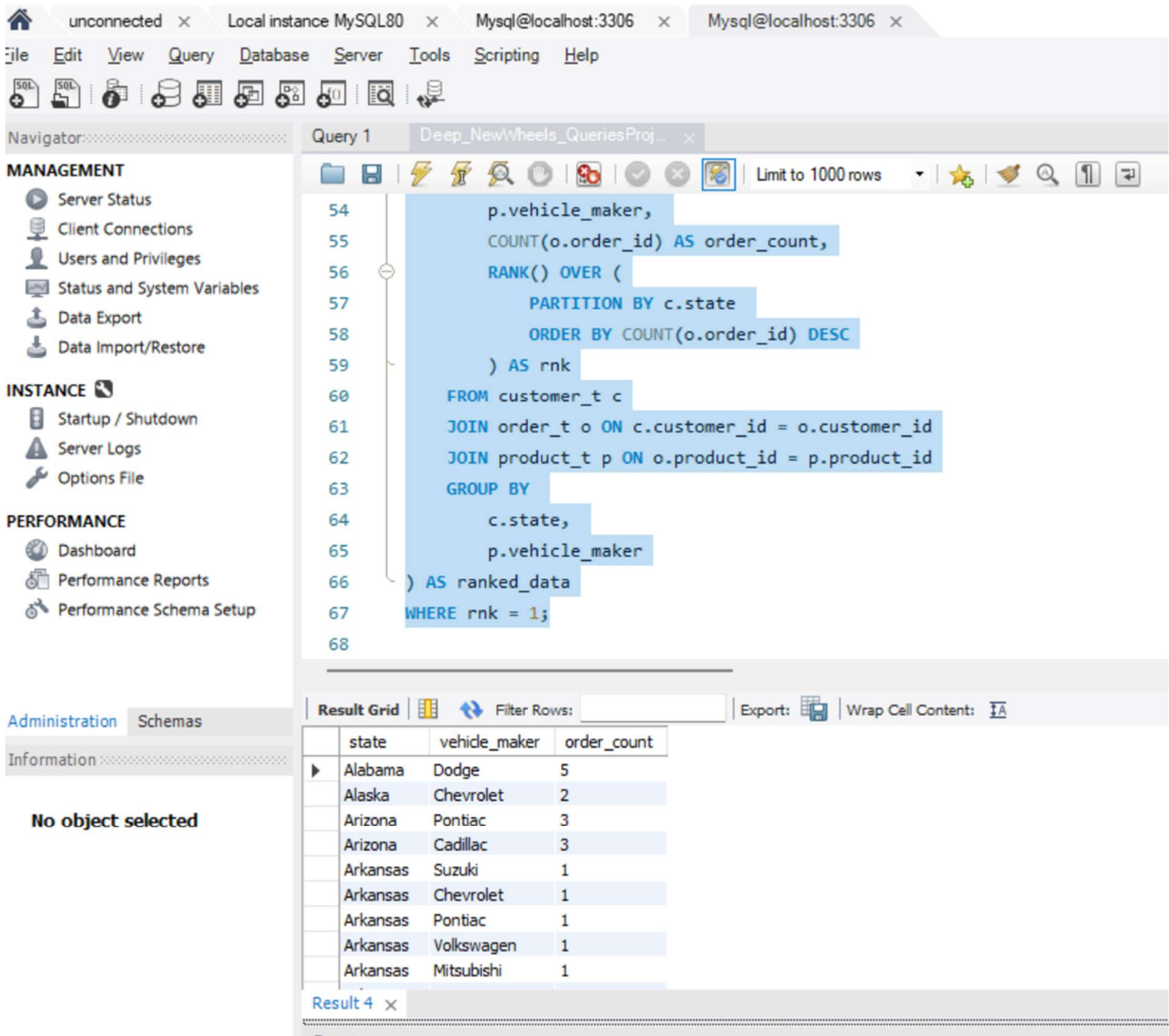
- Query 1
- Query 2
- Query 3

Figure 6 Question 3 Output Screenshot

Showing first 10 rows out of 143 rows

state	vehicle_maker	order_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1

MySQL Workbench



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The main area displays a SQL query in the 'Query 1' tab. The query is as follows:

```

54 p.vehicle_maker,
55 COUNT(o.order_id) AS order_count,
56 RANK() OVER (
57     PARTITION BY c.state
58     ORDER BY COUNT(o.order_id) DESC
59 ) AS rnk
60 FROM customer_t c
61 JOIN order_t o ON c.customer_id = o.customer_id
62 JOIN product_t p ON o.product_id = p.product_id
63 GROUP BY
64     c.state,
65     p.vehicle_maker
66 ) AS ranked_data
67 WHERE rnk = 1;
68

```

Below the query, the 'Result Grid' shows the output of the query. The table has four columns: 'state', 'vehicle_maker', and 'order_count'. The results are as follows:

state	vehicle_maker	order_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Suzuki	1
Arkansas	Chevrolet	1
Arkansas	Pontiac	1
Arkansas	Volkswagen	1
Arkansas	Mitsubishi	1

Figure 7 Question 3 MySQL Screenshot

Table 3 Question 3 Output

state	vehicle_maker	order_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Suzuki	1
Arkansas	Chevrolet	1
Arkansas	Pontiac	1
Arkansas	Volkswagen	1

Arkansas	Mitsubishi	1
Arkansas	GMC	1
California	Ford	6
California	Dodge	6
California	Audi	6
California	Nissan	6
California	Chevrolet	6
Colorado	Chevrolet	5
Connecticut	Chevrolet	2
Connecticut	Mercury	2
Connecticut	Maserati	2
Connecticut	Volvo	2
Delaware	Mitsubishi	2
District of Columbia	Chevrolet	4
Florida	Toyota	7
Georgia	Toyota	3
Hawaii	Ford	1
Hawaii	Toyota	1
Hawaii	Pontiac	1
Hawaii	Nissan	1
Hawaii	Cadillac	1
Hawaii	GMC	1
Idaho	Dodge	2
Illinois	Ford	3
Illinois	GMC	3
Illinois	Chevrolet	3
Indiana	Mazda	4
Iowa	Chrysler	1
Iowa	Chevrolet	1
Iowa	Hyundai	1
Iowa	Isuzu	1
Iowa	Dodge	1
Iowa	Mazda	1
Iowa	Porsche	1
Iowa	Jeep	1
Iowa	Ford	1
Iowa	Pontiac	1
Iowa	Subaru	1
Kansas	GMC	1
Kansas	Lexus	1
Kansas	Buick	1
Kansas	Mercedes-Benz	1
Kansas	Suzuki	1
Kansas	Honda	1
Kansas	Dodge	1

Kansas	Volkswagen	1
Kansas	Ford	1
Kansas	Mazda	1
Kansas	Maserati	1
Kansas	Nissan	1
Kansas	Saab	1
Kentucky	Acura	1
Kentucky	Mercury	1
Kentucky	Audi	1
Kentucky	Ram	1
Kentucky	Volvo	1
Kentucky	Pontiac	1
Kentucky	Nissan	1
Kentucky	Mercedes-Benz	1
Louisiana	BMW	2
Louisiana	Nissan	2
Louisiana	Ford	2
Louisiana	Pontiac	2
Louisiana	Kia	2
Maine	Mercedes-Benz	1
Maryland	Ford	5
Massachusetts	Dodge	2
Massachusetts	Chevrolet	2
Michigan	Ford	3
Minnesota	GMC	3
Mississippi	Dodge	1
Mississippi	Toyota	1
Missouri	Chevrolet	4
Montana	Chevrolet	1
Montana	Mitsubishi	1
Montana	Dodge	1
Nebraska	Chevrolet	1
Nebraska	Mercedes-Benz	1
Nebraska	Volkswagen	1
Nebraska	Nissan	1
Nebraska	Pontiac	1
Nebraska	Toyota	1
Nebraska	Cadillac	1
Nevada	Pontiac	3
New Hampshire	Chrysler	1
New Hampshire	Lincoln	1
New Hampshire	Lexus	1
New Jersey	Mercedes-Benz	2

New Jersey	Hyundai	2
New Mexico	Dodge	2
New York	Toyota	5
New York	Pontiac	5
North Carolina	Volvo	3
North Dakota	Hyundai	1
North Dakota	Ford	1
Ohio	Chevrolet	6
Oklahoma	Toyota	2
Oklahoma	Ferrari	2
Oklahoma	Mazda	2
Oregon	Toyota	2
Pennsylvania	Toyota	3
South Carolina	Acura	1
South Carolina	Buick	1
South Carolina	BMW	1
South Carolina	Kia	1
South Carolina	Mazda	1
South Carolina	Mitsubishi	1
South Carolina	Dodge	1
South Carolina	Jaguar	1
South Carolina	Isuzu	1
Tennessee	Mazda	3
Texas	Chevrolet	9
Utah	Maybach	1
Utah	Volkswagen	1
Utah	Isuzu	1
Utah	Subaru	1
Utah	Lincoln	1
Utah	Chevrolet	1
Utah	Oldsmobile	1
Utah	Pontiac	1
Utah	Dodge	1
Utah	Buick	1
Vermont	Mazda	1
Virginia	Ford	5
Washington	Chevrolet	5
West Virginia	Mercedes-Benz	2
Wisconsin	Pontiac	1
Wisconsin	Chevrolet	1
Wisconsin	Acura	1
Wisconsin	Mazda	1
Wisconsin	Nissan	1
Wisconsin	Cadillac	1
Wisconsin	Dodge	1
Wisconsin	Honda	1

Wyoming	Buick	1
---------	-------	---

Observations and Insights:

- **Chevrolet** is the dominant vehicle maker, appearing as the top choice in the highest number of instances (**54 orders**), followed by **Ford (28)** and **Toyota (25)**.
- The dataset contains 143 records, which implies that in many states, multiple vehicle makers are tied for the number-one rank
- The graph indicates the distribution.

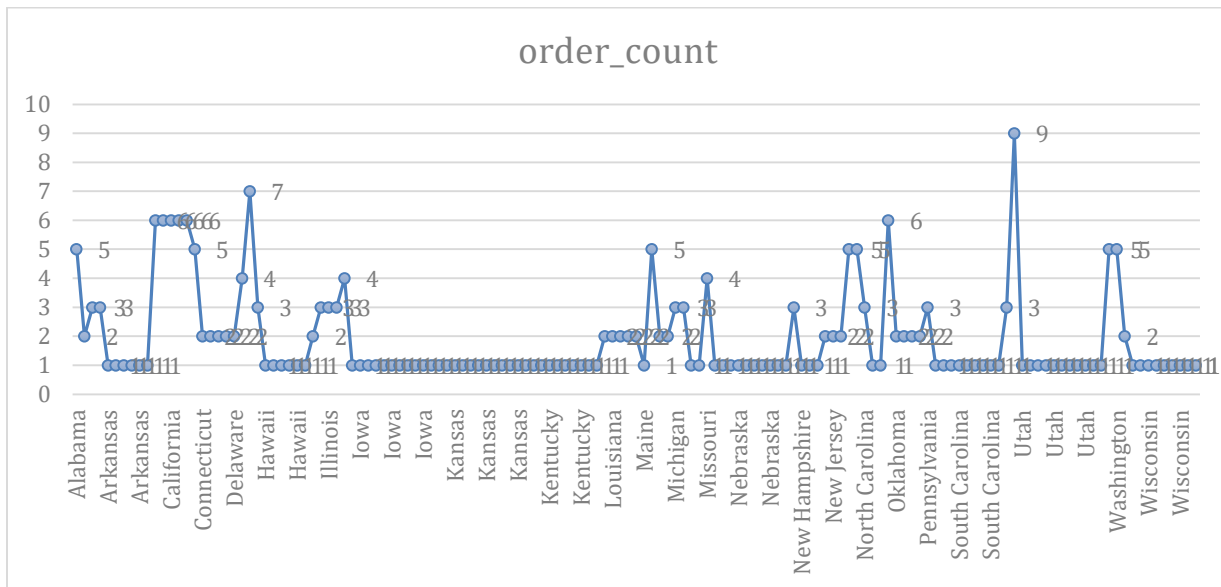


Figure 8 Question 3 Chart

Question 4: Find the overall average rating given by the customers.
What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

--Overall Average Rating

```
SELECT
    quarter_number,
    ROUND(AVG(CASE
        WHEN customer_feedback = 'Very Bad' THEN 1
        WHEN customer_feedback = 'Bad' THEN 2
        WHEN customer_feedback = 'Okay' THEN 3
        WHEN customer_feedback = 'Good' THEN 4
        WHEN customer_feedback = 'Very Good' THEN 5
        ELSE NULL
    END), 2) As rating
FROM order_t;
```

---Average Rating in Each Quarter

```
SELECT
    ROUND(AVG(CASE
        WHEN customer_feedback = 'Very Bad' THEN 1
        WHEN customer_feedback = 'Bad' THEN 2
        WHEN customer_feedback = 'Okay' THEN 3
        WHEN customer_feedback = 'Good' THEN 4
        WHEN customer_feedback = 'Very Good' THEN 5
        ELSE NULL
    END), 2) As rating
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

```

65 -- Hint: Use subquery and assign numerical values to feedback categories using a CASE st
66 -- Then, calculate the average feedback count per quarter. Use a subquery to convert fee
67 -- into numerical values and group by quarter_number to compute the average.
68 --Overall Average Rating
69 SELECT
70     ROUND(AVG(CASE
71         WHEN customer_feedback = 'Very Bad' THEN 1
72         WHEN customer_feedback = 'Bad' THEN 2
73         WHEN customer_feedback = 'Okay' THEN 3
74         WHEN customer_feedback = 'Good' THEN 4
75         WHEN customer_feedback = 'Very Good' THEN 5
76         ELSE NULL
77     END), 2) As rating
78 FROM order_t;
79

```

Run Test

Test Cases

Run SQL

FROM order_t

Output:

Showing 1 rows

rating
3.14


```

79
80 ---Average Rating in Each Quarter
81 SELECT
82     quarter_number,
83     ROUND(AVG(CASE
84         WHEN customer_feedback = 'Very Bad' THEN 1
85         WHEN customer_feedback = 'Bad' THEN 2
86         WHEN customer_feedback = 'Okay' THEN 3
87         WHEN customer_feedback = 'Good' THEN 4
88         WHEN customer_feedback = 'Very Good' THEN 5
89         ELSE NULL
90     END), 2) As rating
91 FROM order_t
92 GROUP BY quarter_number
93 ORDER BY quarter_number;

```

Run SQL

quarter_number	rating
1	3.55
2	3.35
3	2.96
4	2.4

Figure 9 Question 4 Output

rating
3.14

Table 4 Question 4 Output

quarter_number	Rating
1	3.55
2	3.35
3	2.96
4	2.4

Table 2 Question 4 Output Table

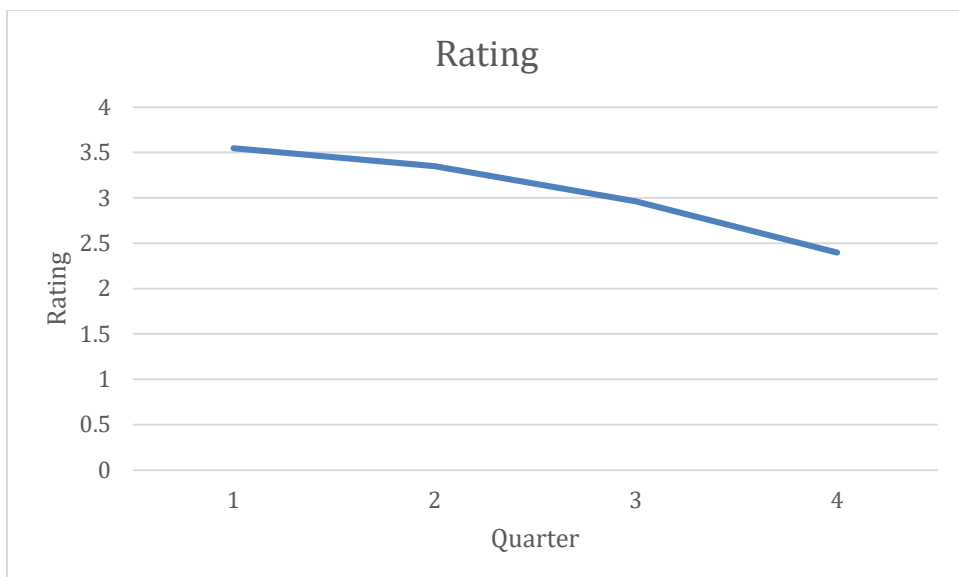


Figure 10 Question 4 Chart

Observations and Insights:

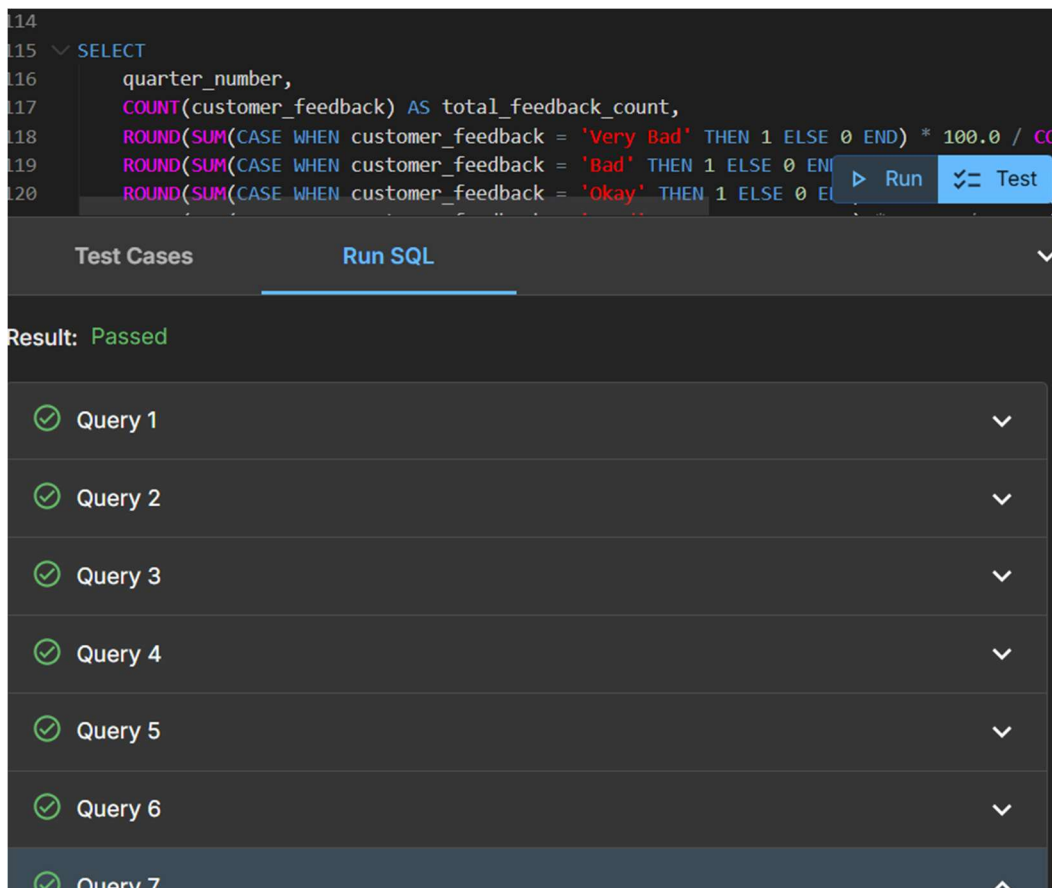
- The overall rating is 3.4
- **Declining Satisfaction:** There is a clear and concerning downward trend in customer satisfaction over the year. Ratings have consistently dropped every quarter, starting at a high of **3.55 (Q1)** and ending at **2.40 (Q4)**
- The rapid acceleration in dissatisfaction during Q3 and Q4 suggests that a specific event, change in process, or systemic issue may have begun mid-year and intensified. It is highly recommended to investigate operational changes

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(customer_feedback) AS total_feedback_count,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS very_bad_percent,
    ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS bad_percent,
    ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS okay_percent,
    ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS good_percent,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS very_good_percent
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL IDE interface. At the top, a SQL query is entered in a text area. Below the query area, there are buttons for 'Run' and 'Test'. The 'Run' button is highlighted. Below the buttons, there is a tabbed interface with 'Test Cases' and 'Run SQL' tabs. The 'Run SQL' tab is active. Below the tabs, the result is displayed as 'Result: Passed'. Below this, there is a list of queries, each with a green checkmark and a dropdown arrow. The queries are labeled 'Query 1' through 'Query 7'.

```

114
115 SELECT
116     quarter_number,
117     COUNT(customer_feedback) AS total_feedback_count,
118     ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback)) AS very_bad_percent,
119     ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback)) AS bad_percent,
120     ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback)) AS okay_percent,
    
```

Run SQL

Test Cases

Output:

Showing 4 rows

quarter_number	total_feedback_count	very_bad_percent	bad_percent	okay_percent	good_percent
1	310	10.97	11.29	19.03	28.71
2	262	14.89	14.12	20.23	22.14
3	229	17.9	22.71	21.83	20.96
4	199	30.65	29.15	20.1	10.05

Figure 11 Question 5 Output Screenshot

Table 5 Question 5 Output

quarter_number	total_feedback_count	very_bad_percent	bad_percent	okay_percent	good_percent	very_good_percent
1	310	10.97	11.29	19.03	28.71	30
2	262	14.89	14.12	20.23	22.14	28.63
3	229	17.9	22.71	21.83	20.96	16.59
4	199	30.65	29.15	20.1	10.05	10.05

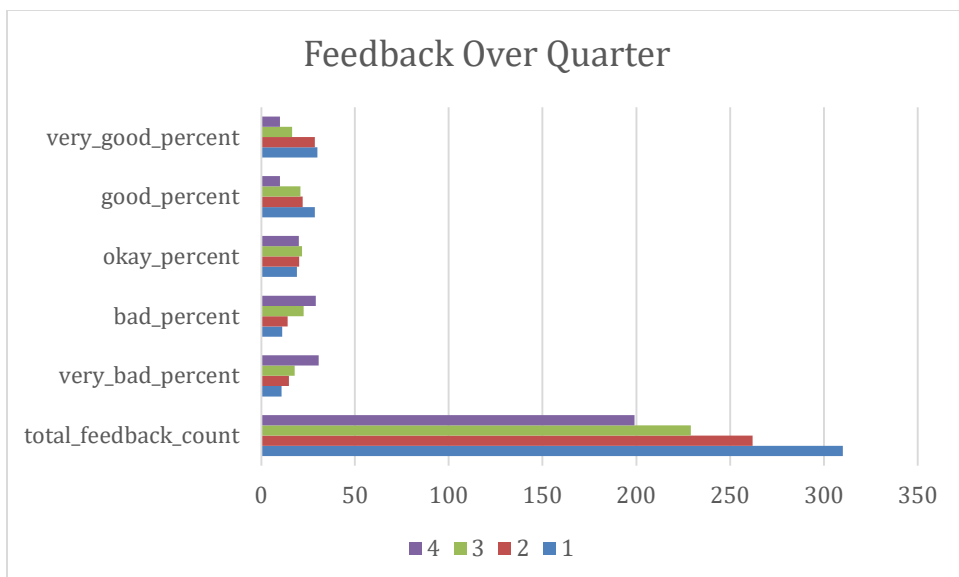


Figure 12 Question 5 Chart

Observations and Insights:

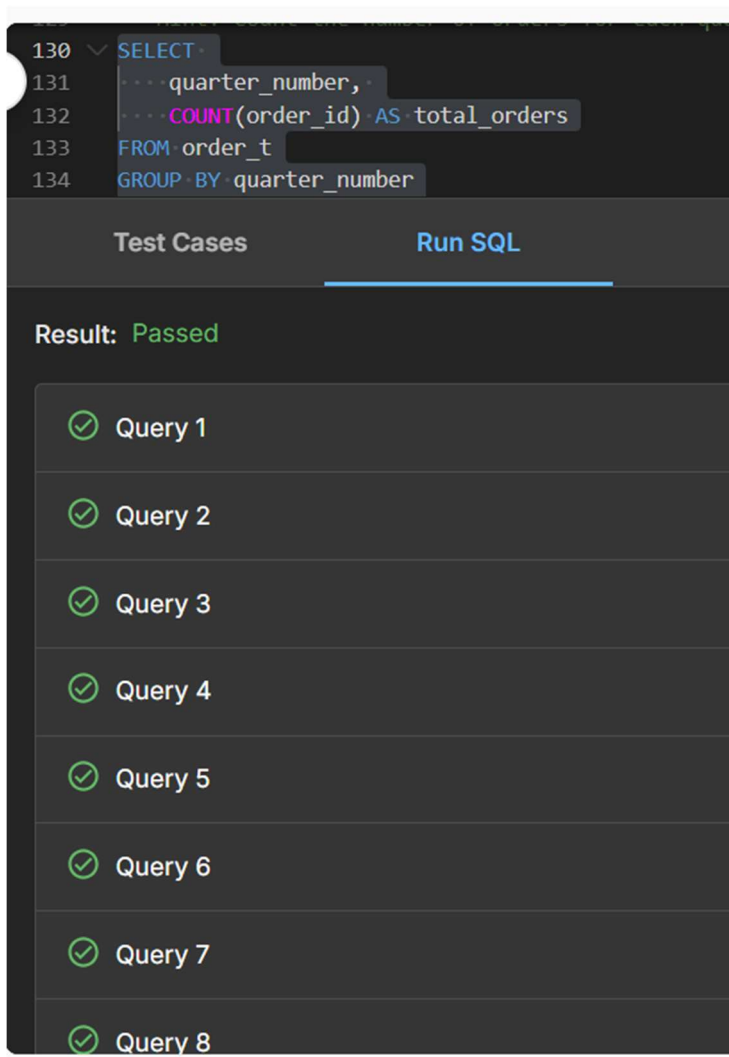
- **Declining Sentiment:** Over the course of the year, customer sentiment shifted dramatically from positive to negative.
- **Escalating Dissatisfaction:** The percentage of customers rating their experience as "Very Bad" almost tripled, rising from **10.97% in Q1** to **30.65% in Q4**. Similarly, "Bad" ratings increased from **11.29% to 29.15%** over the same period.
- **Declining Engagement:** The total volume of feedback received dropped steadily throughout the year, from **310 in Q1** to **199 in Q4**. This decrease in feedback, paired with increasingly negative sentiment, may suggest declining customer loyalty or engagement.
- Interestingly, the "Okay" sentiment remained relatively stable throughout the year, hovering around **19–21%**, suggesting that a core group of customers maintained a moderate experience while those who were previously satisfied shifted into the dissatisfied categories.
- The acceleration of negative sentiment, particularly between Q3 and Q4, indicates that whatever issues were impacting the customer experience likely intensified toward the end of the year.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL query editor with the following SQL code:

```
130 SELECT
131     quarter_number,
132     COUNT(order_id) AS total_orders
133 FROM order_t
134 GROUP BY quarter_number
```

Below the query editor, there are two tabs: "Test Cases" and "Run SQL". The "Run SQL" tab is active, and the result is displayed as "Result: Passed".

Below the result, there is a list of eight queries, each with a green checkmark icon and the text "Query 1" through "Query 8".

Test Cases
Run SQL

```

SELECT
  quarter_number,
  COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number

```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Figure 13 Question 6 Output Screenshot

Table 6 Question 6 Output

quarter_number	total_orders
1	310
2	262
3	229
4	199



Figure 14 Question 6 Chart

Observations and Insights:

- **Consistent Decline in Volume:** There is a steady and significant downward trend in the number of orders placed each quarter throughout the year.
- The total order volume dropped from **310 in Q1** to **199 in Q4**, representing an overall decrease of approximately **35.8%** across the year.
- A consistent, multi-quarter decline is a critical indicator of a struggling business model or operational failure. Immediate steps are needed to identify the root cause of the declining sales

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    quarter_number,
    net_revenue,
    CASE
        WHEN last_quarter_revenue > 0 THEN
            ROUND(((net_revenue - last_quarter_revenue) / last_quarter_revenue) * 100, 2)
        ELSE NULL
    END AS qoq_percentage_change
FROM (
    SELECT
        quarter_number,
        ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue,
        LAG(ROUND(SUM((quantity * vehicle_price) - discount), 2))
            OVER (ORDER BY quarter_number) AS last_quarter_revenue
    FROM order_t
    GROUP BY quarter_number
) AS revenue_table
ORDER BY quarter_number;
```

Output:

Table 7 Question7 Output

net_revenue
125482191.37

quarter_number	net_revenue	qoq_percentage_change
1	39637462.49	NULL
2	32913580.04	-16.96
3	29435267.57	-10.57
4	23495881.27	-20.18

Test Cases

Run SQL

Result: Passed

✓ Query 1

✓ Query 2

✓ Query 3

✓ Query 4

✓ Query 5

✓ Query 6

✓ Query 7

✓ Query 8

✓ Query 9

✓ Query 10

133 SELECT
134 ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue
135 FROM order_t;
136
137 SELECT

▶ Run Test

Test Cases

Run SQL

Result: Passed

```

139 -- Calculate the net revenue. What is the quarter-over-quarter % change? [5 marks]
140 -- Hint: Net Revenue is the amount obtained by multiplying the number of units sold
141 -- by the price after deducting the discounts applied.
142 -- Quarter over Quarter percentage change in revenue means what is the change in revenue
143 -- Calculate the revenue for each quarter by summing the quantity of product and the d
144 -- Ensure the results are ordered by quarter_number to maintain the correct sequence.
145 SELECT
146     quarter_number,
147     net_revenue,
148     CASE
149         WHEN last_quarter_revenue > 0 THEN
150             ROUND(((net_revenue - last_quarter_revenue) / last_qua
151     ELSE NULL

```

Run Test

Test Cases

Run SQL

Showing 4 rows

quarter_number	net_revenue	qoq_percentage_change
1	39637462.49	
2	32913580.04	-16.96
3	29435267.57	-10.57
4	23495881.27	-20.18

Figure 15 Question 7 Output Screenshot

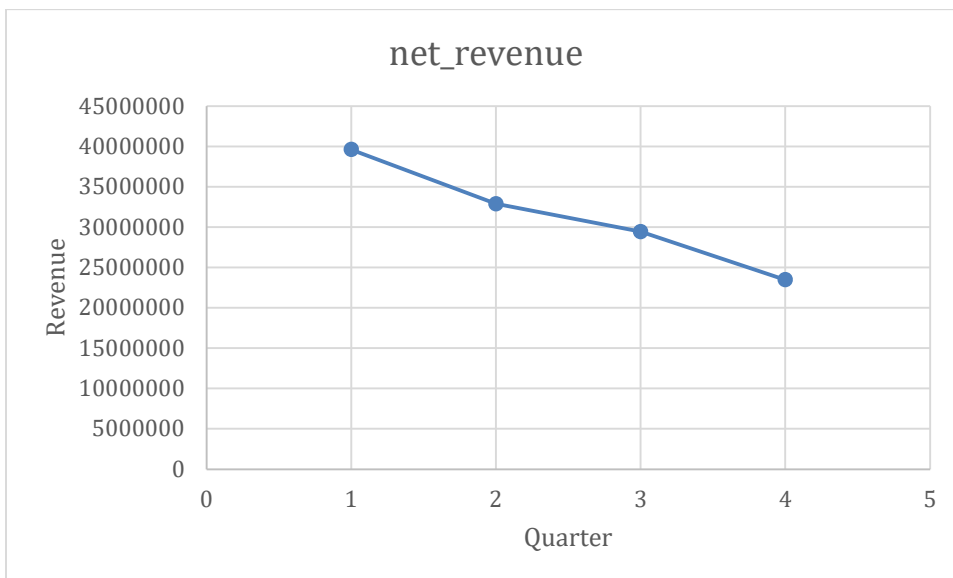
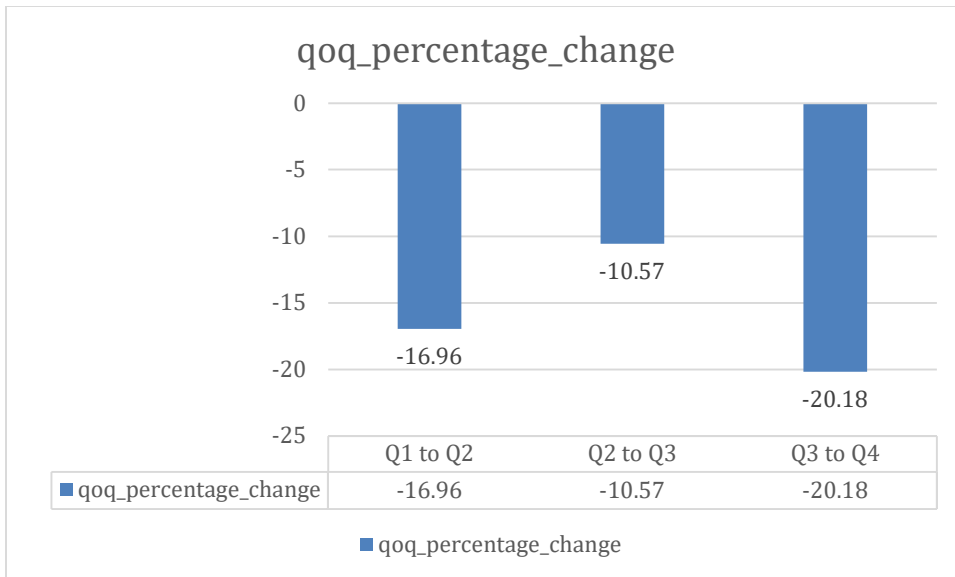


Figure 16 Question 7 Chart



Observations and Insights:

- **Significant Revenue Contraction:** The company experienced a continuous decline in net revenue throughout the year. Total revenue for the year reached **125.48 million**, but the performance gap between Q1 and Q4 is severe.
- **Consistent Downward Trajectory:** The company has failed to stabilize revenue in any quarter. Each subsequent quarter has performed worse than the previous one, confirming that the issues facing the company are persistent and worsening

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```

SELECT
    ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue
FROM order_t;

SELECT
    quarter_number,
    net_revenue,
    last_quarter_revenue,
    CASE
        WHEN last_quarter_revenue > 0 THEN
            ROUND(((net_revenue - last_quarter_revenue) / last_quarter_revenue) * 100, 2)
        ELSE NULL
    END AS revenue_change_percent,
    current_quarter_total_orders,
    last_quarter_total_orders,
    CASE
        WHEN last_quarter_total_orders > 0 THEN
            ROUND(((current_quarter_total_orders - last_quarter_total_orders) /
last_quarter_total_orders) * 100, 2)
        ELSE NULL
    END AS order_change_percent
FROM (
    SELECT
        quarter_number,
        ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue,
        LAG( ROUND(sum(((quantity * vehicle_price) - discount)),2) ) OVER (ORDER BY
quarter_number ) AS last_quarter_revenue,
        COUNT(order_id) AS current_quarter_total_orders,
        LAG ( COUNT(order_id) ) OVER (ORDER BY quarter_number ) AS last_quarter_total_orders
    FROM order_t
    GROUP BY quarter_number
    ORDER BY quarter_number
) AS revenue_order_table
ORDER BY quarter_number;

```

Output:

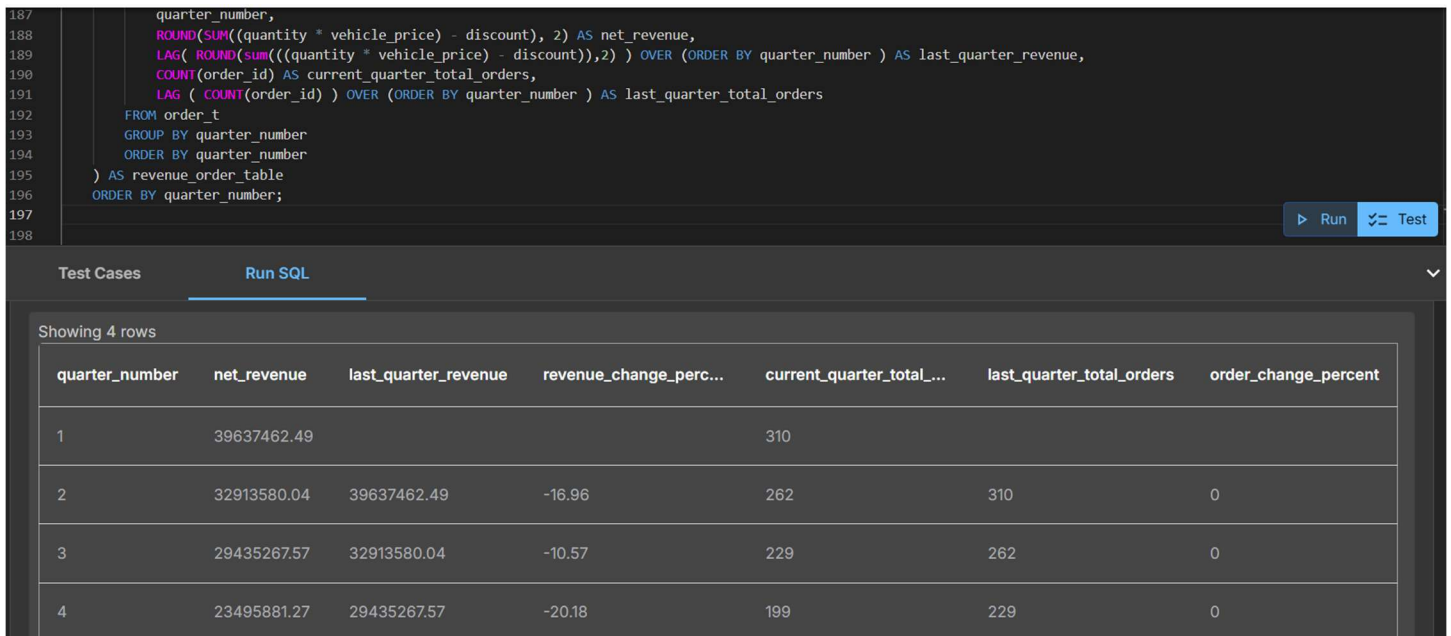
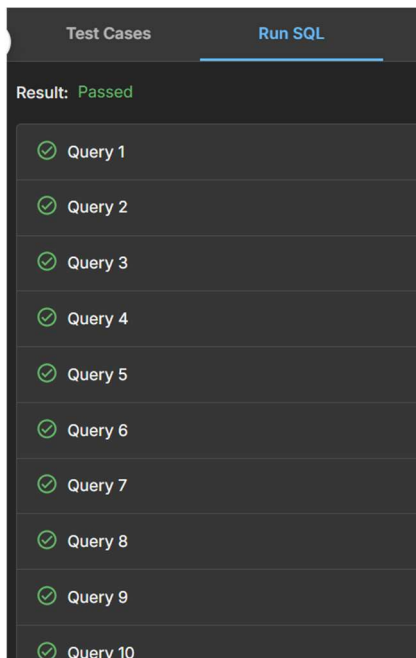


Figure 17 Question 8 Output Screenshot

Table 8 Question 8 Output

quarter_number	net_revenue	last_quarter_revenue	revenue_change_percent	current_quarter_total_orders	last_quarter_total_orders	order_change_percent
1	39637462.49	NULL	NULL	310	NULL	NULL
2	32913580.04	39637462.49	-16.96	262	310	-15.48

3	29435267.57	32913580.04	-10.57	229	262	-12.6
4	23495881.27	29435267.57	-20.18	199	229	-13.1

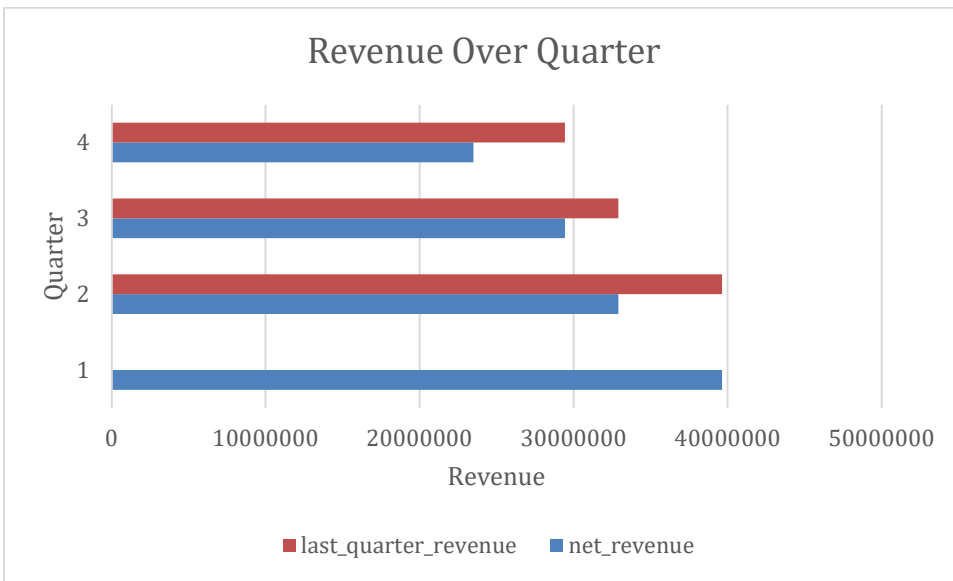


Figure 18 Question 8 Chart



Observations and Insights:

- Both net revenue and order volume are in a sustained downward trend across all four quarters. The order count contracted by **35.8%** from Q1 to Q4, while net revenue dropped by **40.7%** over the same period.
- The decline in total revenue is not just a function of fewer sales (volume). In the final quarter (Q4), the sharper revenue drop (-20.18%) compared to the order drop (-13.1%) indicates that



the company is not only selling fewer units but also selling them at a lower average price or experiencing a shift toward lower-value products.

- **Operational Health:** The business is consistently losing both volume and value. The fact that the revenue loss (-20.18%) in Q4 was significantly worse than the volume loss (-13.1%) suggests that the company's ability to generate value per transaction is eroding.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    credit_card_type,
    ROUND(AVG(discount), 2) AS average_discount
FROM customer_t
JOIN order_t ON customer_t.customer_id = order_t.customer_id
GROUP BY credit_card_type
ORDER BY average_discount DESC;
```

Output:

X
SQL Playground - Project • 1 Question

```

202 -- Hint: Find out the average of discount for each credit card type.
203 SELECT
204     credit_card_type,
205     ROUND(AVG(discount), 2) AS average_discount
206 FROM customer_t
207 JOIN order_t ON customer_t.customer_id = order_t.customer_id
208 GROUP BY credit_card_type
209 ORDER BY average_discount DESC;
210
211

```

Test Cases
Run SQL

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62

Figure 19 Question 9 Output Screenshot

Query 1 Deep_NewWheels_QueriesProj... x

Limit to 1000 rows

```

189 • SELECT
190     credit_card_type,
191     ROUND(AVG(discount), 2) AS average_discount
192 FROM customer_t
193 JOIN order_t ON customer_t.customer_id = order_t.customer_id
194 GROUP BY credit_card_type
195 ORDER BY average_discount DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: ☐

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62
maestro	0.62
instapayment	0.62
jcb	0.61
switch	0.61
diners-club-carte-blanche	0.61
bankcard	0.61
diners-club-us-ca	0.61
diners-club-enroute	0.60
visa	0.60
solo	0.59
diners-club-international	0.58

Figure 20 Question 9 MySQL Screenshot

Table 9 Question 9 Output

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62
maestro	0.62
instapayment	0.62
jcb	0.61
switch	0.61
diners-club-carte-blanche	0.61
bankcard	0.61

diners-club-us-ca	0.61
diners-club-enroute	0.6
visa	0.6
solo	0.59
diners-club-international	0.58

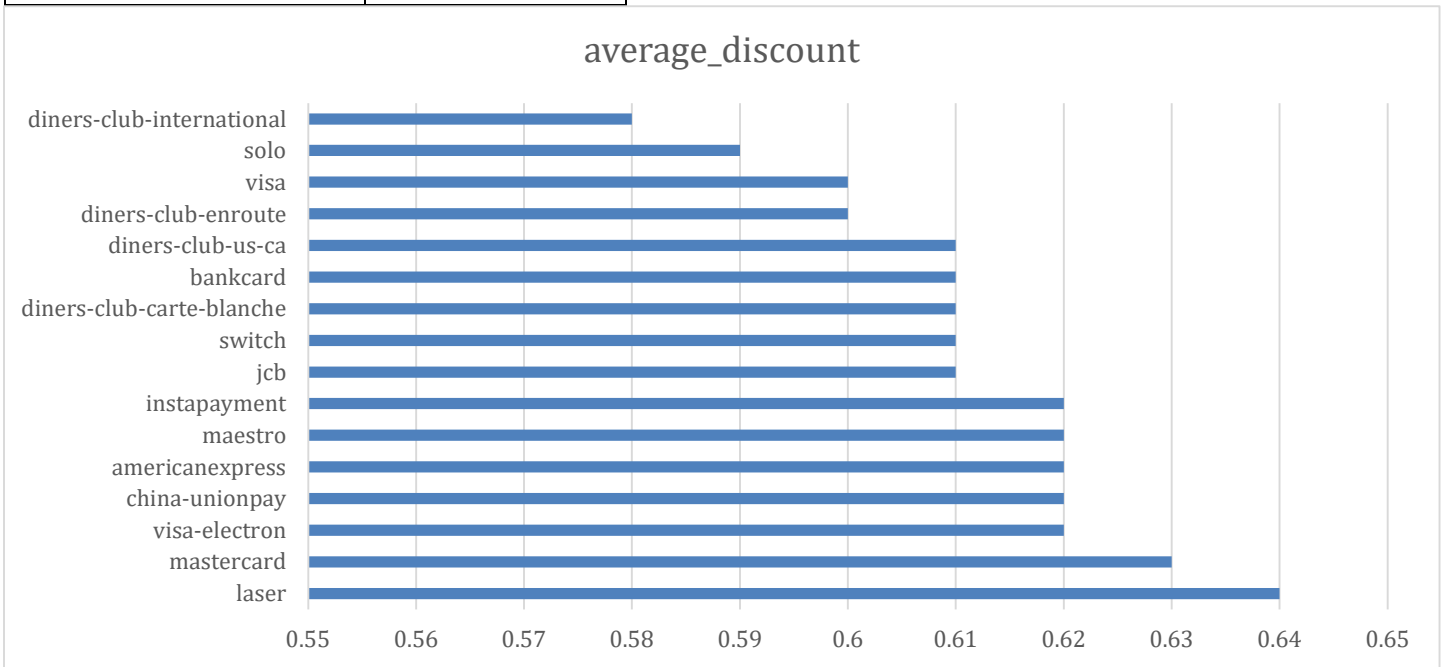


Figure 21Question 9 Chart

Observations and Insights:

- **High Consistency:** The average discounts offered across different credit card types are remarkably uniform, ranging only from **0.58 to 0.64**. The mean discount is approximately **0.61**, with a very low standard deviation (**~0.015**), indicating that customers receive roughly the same discount regardless of the payment method they choose.
- While the range is small, laser (0.64) and mastercard (0.63) sit at the top end of the discount spectrum, while diners-club-international (0.58) and solo (0.59) are at the bottom.
- The data suggests that your pricing strategy is not currently using discounts as a tool to incentivize the use of specific credit card types. This is a neutral finding, as it implies your current policy treats all credit card customers equitably regarding discounts.
- If the goal was to drive volume to specific, lower-fee payment methods, this data indicates that the current discount policy is not configured to achieve that outcome.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

--SQLite

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS average_shipping_time
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

--MySQL

```
SELECT quarter_number,
    ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS average_shipping_time
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number ;
```

Output:

Table 10 Question 10 Output

quarter_number	average_shipping_time
1	57.17
2	71.11
3	117.76
4	174.1

SQL Playground - Project • 1 Question

```

218
219
220 SELECT
221     quarter_number,
222     ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS average_shipping_time
223 FROM order_t
224 GROUP BY quarter_number
225 ORDER BY quarter_number;
  
```

Test Cases Run SQL

Output:

Showing 4 rows

quarter_number	average_shipping_time
1	57.17
2	71.11
3	117.76
4	174.1

Figure 22 Question 10 SQLite Output Screenshot

MySQL Workbench

Mysql@localhost:3306 x

File Edit View Query Database Server Tools Scripting Help

Navigator: MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Query 1 Deep_NewWheels_QueriesProj...

```

197 SELECT quarter_number,
198     ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS average_shipping_time
199 FROM order_t
200 GROUP BY quarter_number
201 ORDER BY quarter_number ;
202
203
  
```

Limit to 1000 rows

Result Grid

quarter_	average_shipping_time
1	57.17
2	71.11
3	117.76
4	174.10

Figure 23 Question 10 MySQL Output Screenshot

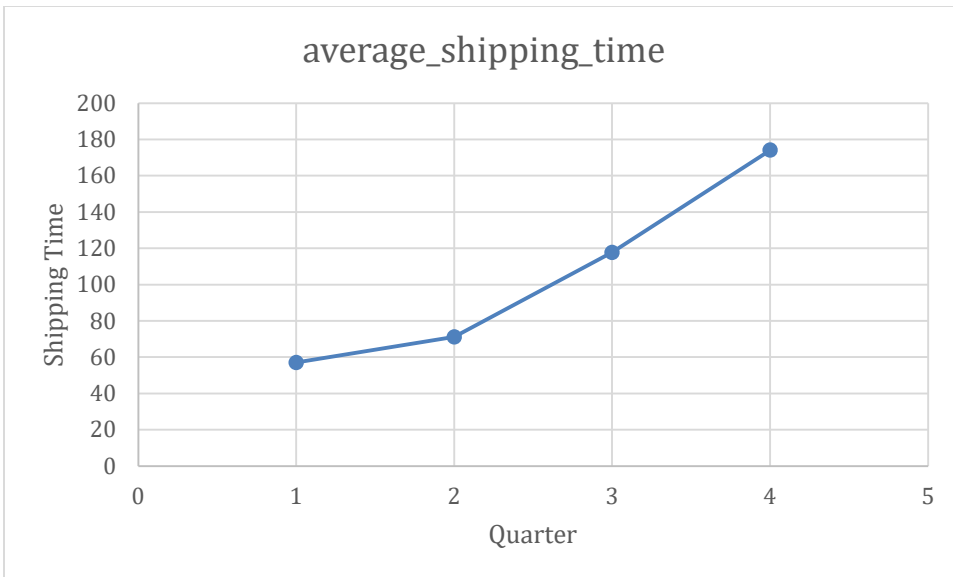


Figure 24 Question 10 Chart

Observations and Insights:

- **Rapidly Increasing Shipping Times:** There is a sharp, consistent upward trend in the average time taken to ship orders.
- **Exponential Growth in Delays:** The delays grew progressively worse each quarter. The increase accelerated dramatically in Q3, where shipping time grew by over 65% compared to Q2. Overall, the average shipping time in Q4 was more than three times longer than it was in Q1.
- **Operational Instability:** The jump from a ~57-day shipping average to a ~174-day average is an extremely significant operational failure. This indicates a complete breakdown in logistics, supply chain management, or fulfillment capacity as the year progressed.
- **Critical Risk:** Shipping delays of nearly half a year (in Q4) are unacceptable for most industries and are likely causing irreparable damage to brand trust

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
125482191.37	1000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
23495881.27	229	97.96	21.5%

Queries Used for Deriving Business Metrics:

```
SELECT
    ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue,
    COUNT(order_id) AS total_orders
FROM order_t;
```

Table 11 Business Metrics Query 1 -Net Revenue and Total Orders

net_revenue	total_orders
125482191.37	1000

```
SELECT
    COUNT(DISTINCT customer_id) AS total_customers
FROM customer_t;
```

Table 12 Business Metrics Query 2 – Total Customers

total_customers
994

```
SELECT
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS average_shipping_time
FROM order_t;
```

Table 13 Business Metrics Query 3 – Average Shipping Time

average_shipping_time
97.96

```
SELECT
SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS very_bad_count,
SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) AS bad_count,
SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) AS okay_count,
SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) AS good_count,
SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS very_good_count
FROM order_t;
```

Table 14 Business Metrics Query 4 – Feedback Count

very_bad_count	bad_count	okay_count	good_count	very_good_count
175	182	202	215	226

```
SELECT
COUNT(customer_feedback) AS total_feedback_count,
ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS very_bad_percent,
ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS bad_percent,
ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS okay_percent,
ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS good_percent,
ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_feedback), 2) AS very_good_percent
FROM order_t;
```

Table 15 Business Metrics Query 5 – Feedback Percentage

total_feedback_count	very_bad_percent	bad_percent	okay_percent	good_percent	very_good_percent
1000	17.5	18.2	20.2	21.5	22.6

Evidence For Running Queries:


```

227 -- For Business Metrics
228 /* Business Metrics */
229 SELECT
230     ROUND(SUM((quantity * vehicle_price) - discount), 2) AS net_revenue,
231     COUNT(order_id) AS total_orders
232 FROM order_t;
233
234 SELECT
235     COUNT(DISTINCT customer_id) AS total_customers
236 FROM customer_t;
237
238 SELECT
239     ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS average_shipping_time
240 FROM order_t;
241
242 SELECT
243     SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS very_bad_count,
244     SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) AS good_count,
245     SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS very_good_count,
246     SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) AS bad_count
247 FROM customer_t;

```

Test Cases

Run SQL

✓ Query 14

✓ Query 15

✓ Query 16

Query:

Figure 25 Business Metrics Evidence

Insights

- Annual Performance Metrics:** The company generated a total annual revenue of **125,482,191.37** from **1,000 total orders** placed by **994 unique customers**. The overall customer satisfaction, measured as an average rating, is **3.14** on a 5-point scale.
- Current Quarterly Status:** In the final quarter, revenue was **23,495,881.27** across **229 orders**. The logistical performance in this period was marked by a high average shipping time of **97.96 days**, while customer sentiment regarding "Good" feedback stood at **21.5%**.
- Severe Operational Decline:** The company is experiencing a critical year-end crisis, evidenced by a 40.72% drop in revenue and a 35.8% decrease in order volume from Q1 to Q4.
- Logistical Breakdown:** Average shipping times have deteriorated exponentially, ballooning from 57.17 days in Q1 to 174.1 days in Q4, which is the primary driver for the parallel collapse in customer sentiment.
- Collapse of Customer Trust:** Positive customer feedback ("Good" or "Very Good") plummeted from 58.71% in Q1 to a mere 20.10% in Q4, reflecting a deep-seated dissatisfaction that threatens long-term brand viability.

Business Recommendations

- **Conduct an Immediate Supply Chain Audit:** Investigate the root cause of the shipping delay crisis, focusing on third-party logistics partners, warehouse bottlenecks, and potential inventory shortages.
- **Prioritize Customer Recovery:** Launch a proactive customer communication plan for those affected by Q3 and Q4 delays, offering explicit apologies, transparent timelines, and compensation to rebuild damaged trust.
- **Re-evaluate Pricing and Discount Strategy:** Analyse if aggressive discounting, which has resulted in declining Average Revenue Per Order (ARPO), is actually harming profitability and perceived product value.
- **Optimize Product Portfolio:** Review the Q4 revenue collapse against the product mix to determine if low-margin items are diluting sales volume or if core high-value products are suffering from availability issues.
- **Strengthen Quality Control:** Since negative feedback ("Bad"/"Very Bad") now accounts for nearly 60% of Q4 sentiment, implement stricter quality assurance protocols to ensure product deliveries meet customer expectations once fulfilment speeds are stabilized.
- **Restructure Sales & Marketing:** Pivot marketing spend away from new customer acquisition, which is currently inefficient, and redirect resources toward retaining existing customers with high lifetime value potential.